

1. Find the absolute extrema of $g(x) = 3x^4 - 26x^3 + 60x^2 - 11$ on the interval $[1, 5]$.

$$g(x) = 3x^4 - 26x^3 + 60x^2 - 11$$

$$g'(x) = 12x^3 - 78x^2 + 120x$$

$$= 6x(2x^2 - 13x + 20)$$

$$= 6x(x-4)(2x-5)$$

$$x = 0, 4, \frac{5}{2}$$

∴

$$f(1) = 26 \rightarrow \min$$

$$f(2.5) = 79.9375$$

$$f(4) = 53$$

$$f(5) = 114 \rightarrow \max$$

∴ absolute max @ $(5, 114)$

absolute min @ $(1, 26)$

2. Determine the equation of the tangent to the curve $3y^2 + 6xy + 4x^2 - 2y = 5$ at the point $(-2, 1)$. Use exact values.

$$3y^2 + 6xy + 4x^2 - 2y = 5 \text{ @ } (-2, 1)$$

$$6y \frac{dy}{dx} + 6x \frac{dy}{dx} + 6y + 8x - 2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-6y - 8x}{6y + 6x - 2}$$

$$\frac{dy}{dx} = \frac{-6(1) - 8(-2)}{6(1) + 6(-2) - 2}$$

$$= -\frac{5}{4}$$

$$1 = -\frac{5}{4}(-2) + b$$

$$y = -\frac{5}{4}x - \frac{3}{2}$$

3. A paper airplane is thrown off the roof of the building. The position function, $s(t)$, in metres, of the paper airplane above the ground is $s(t) = -2t^3 - 15t^2 + 36t + 10$, where t is in seconds, $t \geq 0$.

- a) Determine when the maximum height is reached by the paper airplane.

$$s(t) = -2t^3 - 15t^2 + 36t + 10$$

$$v(t) = -6t^2 - 30t + 36$$

$$= 6(-t^2 - 5t + 6)$$

- b) Determine the time at which the paper airplane will have an acceleration of -48 m/s^2 .

$$a(t) = -12t - 30$$

$$-48 = -12t - 30$$

$$-18 = -12t$$

@ $3/2$ seconds

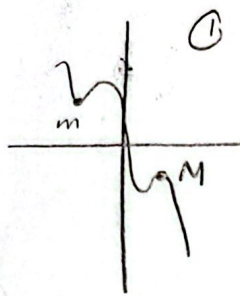
- c) What is the height of the airplane at the time found in part b)?

$$s(t) = -2\left(\frac{3}{2}\right)^3 - 15\left(\frac{3}{2}\right)^2 + 36\left(\frac{3}{2}\right) + 10$$

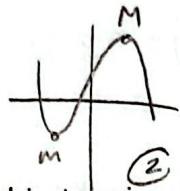
$$= \frac{47}{2} \rightarrow 23.5 \text{ m}$$

∴ @ $\frac{3}{2}$ seconds height will be 23.5 m

4. If m is a local minimum and M is a local maximum of a continuous function f on the interval (a, b) , then $m < M$. Include a diagram as part of your explanation.



FALSE, CAN BE TRUE SOMETIMES BUT NOT ALWAYS. In case 1, it can be seen how despite m is a local min and M is a local max, $m > M$. But like case 2, M can $> m$ in certain functions.



5. The position of an object moving along a straight line is given by $s(t) = 2t^4 - 16t^3 + 20t^2 - 7$, where $s(t)$ is the object's position in meters after t seconds for $0 \leq t \leq 6$.

- a) When does the object change directions? Justify your answer.

$$v(t) = 8t^3 - 48t^2 + 40t = 8t(t^2 - 6t + 5)$$

$$t = 0, 5, 1$$

changes direction @ 1 sec and 5 sec

$[0, 1) (1, 5) (5, 6]$
 $v(t) (+) (-) (+)$

- b) What is the acceleration of the object at 3 seconds?

$$a(t) = 24t^2 - 96t + 40$$

$$a(3) = -32 \text{ m/s}^2$$

@ 3 sec, acceleration = -32 m/s^2

- c) Is the object speeding up or slowing down at 2 seconds?

$$v(2) = -48 \text{ m/s} \rightarrow (-)$$

$$a(2) = -56 \text{ m/s}^2 \rightarrow (-)$$

same sign

$\therefore$ object is speeding up @ 2 sec.

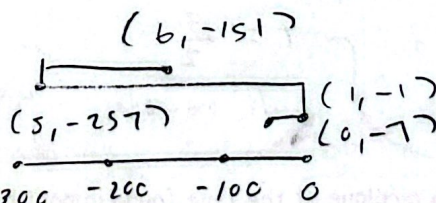
- d) Determine the total distance travelled by the object from $0 \leq t \leq 6$.

$$s(0) = -7 \text{ m}$$

$$s(1) = -1 \text{ m}$$

$$s(5) = -257 \text{ m}$$

$$s(6) = -151 \text{ m}$$



$\therefore$ total distance = 368 m

$$\begin{aligned} & \text{①} \rightarrow \text{②} & \text{②} \rightarrow \text{③} & \text{③} \rightarrow \text{④} \\ & = -7 - (-1) & = -1 - (-257) & = -257 - (-151) \\ & = |-6 \text{ m}| & = |256 \text{ m}| & = |-106 \text{ m}| \end{aligned}$$